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W3C RECOMMENDATION • 30 APRIL 2013

PROV-DM: The PROV Data Model

W3C Provenance Working Group Specification (REC-prov-dm-20130430)

This version: http://www.w3.org/TR/2013/REC-prov-dm-20130430/
Latest published: http://www.w3.org/TR/prov-dm/
Implementation report: http://www.w3.org/TR/2013/NOTE-prov-implementations-20130430/
Previous version: http://www.w3.org/TR/2013/PR-prov-dm-20130312/
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ABSTRACT

Provenance is information about entities, activities, and people involved in producing a piece of data or thing, which can be used to form assessments about its quality, reliability or trustworthiness. **PROV-DM** is the conceptual data model that forms a basis for the W3C provenance (PROV) family of specifications. PROV-DM distinguishes *core structures*, forming the essence of provenance information, from *extended structures* catering for more specific uses of provenance. PROV-DM is organized in six components, respectively dealing with: (1) entities and activities, and the time at which they were created, used, or ended; (2) derivations of entities from entities; (3) agents bearing responsibility for entities that were generated and activities that happened; (4) a notion of bundle, a mechanism to support provenance of provenance; (5) properties to link entities that refer to the same thing; and, (6) collections forming a logical structure for its members. This document introduces the provenance concepts found in PROV and defines PROV-DM types and relations. The PROV data model is domain-agnostic, but is equipped with extensibility points allowing domain-specific information to be included.

Status of This Document & PROV Family of Specifications

This document has been reviewed by W3C Members, by software developers, and by other W3C groups and interested parties, and is endorsed by the Director as a **W3C Recommendation**. It is a stable document and may be used as reference material or cited from another document. Two companion Recommendations complete the specification of PROV-DM: **PROV-CONSTRAINTS** (specifying formal constraints for valid provenance) and **PROV-N** (a notation for provenance aimed at human consumption). Additional specifications in the W3C PROV family include: **PROV-OVERVIEW** (Note), **PROV-PRIMER** (Note), **PROV-O** (OWL2 Ontology Recommendation), **PROV-XML** (XML Schema Note), **PROV-AQ** (Access & Query Note), **PROV-DICTIONARY** (Note), **PROV-DC** (Dublin Core Mapping Note), **PROV-SEM** (First-Order Logic Semantics Note), and **PROV-LINKS** (Linking Across Bundles Note).

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1. Introduction, Compliance, and Namespaces

For the purpose of this specification, **provenance** is defined as a record that describes the people, institutions, entities, and activities involved in producing, influencing, or delivering a piece of data or a thing. In particular, the provenance of information is crucial in deciding whether information is to be trusted, how it should be integrated with other diverse information sources, and how to give credit to its originators when reusing it. PROV-DM is a generic, domain-agnostic data model that allows heterogeneous systems to export their native provenance and interchange it across the Web.

1.1 Compliance & 1.3 Notational Conventions (Normative): The normative sections of this specification are Section 1.1, Section 1.3, Section 5, and Appendix A. Information in tables is normative if it appears in a normative section; all UML diagrams and boxes labeled "Example" are informative. The key words "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "MAY", and "OPTIONAL" are to be interpreted as described in `[RFC2119]`. Examples throughout this document use the PROV-N Provenance Notation.

Prefix	Namespace IRI	Definition & Scope (Table 1)
prov	`http://www.w3.org/ns/prov#`	The PROV namespace defining normative PROV-DM terms (Section 5.7.4).
xsd	`http://www.w3.org/2000/10/XMLSchema#`	XML Schema Namespace `[XMLSCHEMA11-2]` for primitive datatypes.
rdf	`http://www.w3.org/1999/02/22-rdf-syntax-ns#`	The RDF namespace `[RDF-CONCEPTS]`.
(others)	`(various, e.g., http://example.com/)`	All other namespace prefixes are used in informative examples only.

2. PROV Overview: Core and Extended Structures

2.1 PROV Core Structures: At its core, provenance relates three fundamental types—**Entity**, **Activity**, and **Agent**—connected by seven core binary relations. Entities represent physical, digital, or conceptual things with fixed aspects. Activities are dynamic processes that occur over a period of time and act upon or with entities. Agents bear responsibility for activities taking place and for the existence of entities.

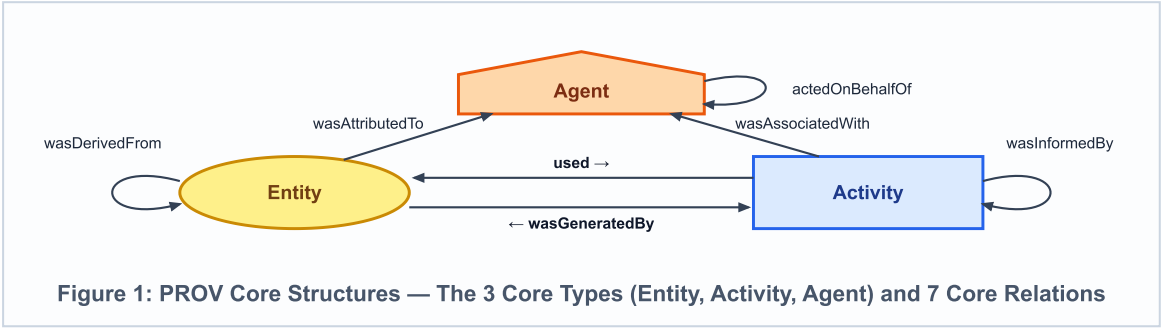


Figure 1: PROV Core Structures — The 3 Core Types (Entity, Activity, Agent) and 7 Core Relations

2.2 PROV Extended Structures & 2.3 Modular Organization: Extended structures enhance the core via four mechanisms: (1) *Subtyping* (e.g., `Person`, `Organization`, and `SoftwareAgent` as subtypes of `Agent`; `Plan`, `Collection`, and `Bundle` as subtypes of `Entity`; `Revision`, `Quotation`, and `PrimarySource` as subtypes of `Derivation`); (2) *Expanded Relations* allowing additional arguments such as timestamps, roles, or plans; (3) *Optional Identification* of relations and parameters using `;` and `-` in PROV-N; and (4) *Further Relations* covering start, end, invalidation, specialization, alternate, and collection membership.

3. The Provenance Notation (PROV-N) & 4. Illustrative Example

PROV-DM examples are expressed in the Provenance Notation (**PROV-N**). Expressions follow a functional syntax: `pred(id; arg_1, ..., arg_n, [attr_1=val_1, ...])`, where `id;` is an optional relation identifier, positional arguments represent core parameters (with `-` denoting omitted optional arguments), and bracketed pairs provide extensible attribute-value metadata.

Section 4 Summary (Provenance of a Technical Report): Section 4 illustrates PROV-DM across three complementary viewpoints: (4.1) *The Authors View*, expressing how document `tr:WD-prov-dm-20111215` was edited and derived from an earlier version `tr:WD-prov-dm-20111018`; (4.2) *The Process View*, detailing the W3C publication activity, publication request email, and publication policy plan; and (4.3) *Attribution of Provenance*, packaging each viewpoint inside a self-contained `bundle` so the provenance record itself can be tracked.

```
      document prefix ex <http://example.org/> prefix tr <http://www.w3.org/TR/2011/>
entity(tr:WD-prov-dm-20111215, [prov:type="document", ex:version="2"]) activity(ex:edit1,
2011-11-15T10:00:00, 2011-12-14T18:00:00, [prov:type="editing"]) wasGeneratedBy(tr:WD-prov-
dm-20111215, ex:edit1, -) wasDerivedFrom(tr:WD-prov-dm-20111215, tr:WD-prov-dm-20111018,
[prov:type='prov:Revision']) agent(ex:Luc, [prov:type='prov:Person']) wasAttributedTo(tr:WD-
prov-dm-20111215, ex:Luc) wasAssociatedWith(ex:edit1, ex:Luc, -, [prov:role="editor"])
endDocument
```

5. PROV-DM Types and Relations (Normative Specification)

Component (Table 5 Index)	Core Structures (Types & Relations)	Extended Structures (Subtypes & Relations)
COMPONENT 1 Entities & Activities	`Entity`, `Activity`, `Generation`, `Usage`, `Communication`	`Start`, `End`, `Invalidation`
COMPONENT 2 Derivations	`Derivation` (`wasDerivedFrom`)	`Revision` (`prov:Revision`), `Quotation` (`prov:Quotation`), `PrimarySource` (`prov:PrimarySource`)
COMPONENT 3 Agents & Responsibility	`Agent`, `Attribution`, `Association`, `Delegation`	`Plan`, `Person`, `Organization`, `SoftwareAgent`, `Influence` (`wasInfluencedBy`)
COMPONENT 4 Bundles	*(None — Extended feature)*	`Bundle constructor` (`bundle id ... endBundle`), `Bundle Type` (`prov:Bundle`)
COMPONENT 5 Alternate Entities	*(None — Extended feature)*	`Specialization` (`specializationOf`), `Alternate` (`alternateOf`)
COMPONENT 6 Collections	*(None — Extended feature)*	`Collection` (`prov:Collection`), `EmptyCollection` (`prov:EmptyCollection`), `Membership` (`hadMember`)

5.1 Component 1: Entities and Activities (Core & Lifecycle Concepts)

5.1.1 Entity — `entity(id, [attrs])`: A physical, digital, conceptual, or other kind of thing with some fixed aspects; entities may be real or imaginary.

5.1.2 Activity — `activity(id, st, et, [attrs])`: Something that occurs over a period of time (from optional start time `st` to end time `et`) and acts upon or with entities; it may include consuming, processing, transforming, modifying, relocating, using, or generating entities.

5.1.3 Generation — `wasGeneratedBy(id; e, a, t, attrs)`: The completion of production of a new entity `e` by an activity `a` at optional time `t`. The entity did not exist before generation and becomes available for usage after this generation.

5.1.4 Usage — `used(id; a, e, t, attrs)`: The beginning of utilizing an entity `e` by an activity `a` at optional time `t`. Before usage, the activity had not begun to utilize this entity and could not have been affected by the entity.

5.1 Component 1 (Continued): Extended Lifecycle Relations

5.1.6 Start — `wasStartedBy(id; a2, e, a1, t, attrs)`: When an activity `a2` is deemed to have been started by an entity `e` (known as the *trigger*), generated by an optional starter activity `a1`, at time `t`. The activity did not exist before its start.

5.1.7 End — `wasEndedBy(id; a2, e, a1, t, attrs)`: When an activity `a2` is deemed to have been ended by an entity `e` (known as the *trigger*), generated by an optional ender activity `a1`, at time `t`. The activity no longer exists after its end.

5.1.8 Invalidation — `wasInvalidatedBy(id; e, a, t, attrs)`: The start of the destruction, cessation, or expiry of an existing entity `e` by an activity `a` at time `t`. The entity is no longer available for use (or further invalidation) after invalidation.

5.2 Component 2: Derivations (Entity-to-Entity Transformation)

The second component of PROV-DM addresses derivations of entities from other entities and defines three specific derivation subtypes represented via the `prov:type` attribute:

5.2.1 Derivation — `wasDerivedFrom(id; e2, e1, a, g2, u1, attrs)`: A transformation of an entity into another, an update of an entity resulting in a new one, or the construction of a new entity based on a pre-existing entity. Here `e2` is the generated entity, `e1` is the used entity, `a` is the underpinning activity, `g2` is the generation identifier, and `u1` is the usage identifier.

5.2.2 Revision — `[prov:type='prov:Revision']`: A derivation for which the resulting entity is a revised version of some original. The implication here is that the resulting entity contains substantial content from the original.

5.2.3 Quotation — `[prov:type='prov:Quotation']`: The repeat of (some or all of) an entity, such as text or image, by someone who may or may not be its original author. Quotation is a particular case of derivation.

5.2.4 Primary Source — `[prov:type='prov:PrimarySource']`: Refers to something produced by some agent with direct experience and knowledge about the topic, at the time of the topic's study, without benefit from hindsight. Because of the directness of primary sources, they "speak for themselves" in ways that cannot be captured through the filter of secondary sources.

5.3 Component 3: Agents, Responsibility, and Influence

The third component concerns agents, their subtypes, responsibility relations linking agents to entities/activities/agents, and the generalized superclass relation of influence:

5.3.1 Agent — `agent(id, [attrs])`: Something that bears some form of responsibility for an activity taking place, for the existence of an entity, or for another agent's activity. PROV-DM defines three normative agent subtypes: `prov:Person` (human agents), `prov:Organization` (social or legal institutions), and `prov:SoftwareAgent` (running software). An entity, activity, or agent may be asserted to have multiple types simultaneously.

5.3.2 Attribution — `wasAttributedTo(id; e, ag, attrs)`: The ascribing of an entity `e` to an agent `ag`. When an entity is attributed to an agent, this entity was generated by some unspecified activity that in turn was associated to that agent.

5.3.3 Association — `wasAssociatedWith(id; a, ag, pl, attrs)`: An assignment of responsibility to an agent `ag` for an activity `a`, indicating that the agent had a role in the activity. It optionally references a **Plan** `pl` (an entity subtype `prov:Plan` representing a set of actions or steps intended by one or more agents to achieve some goal).

5.3.4 Delegation — `actedOnBehalfOf(id; ag2, ag1, a, attrs)`: The assignment of authority and responsibility to an agent `ag2` (subordinate/delegate) to carry out a specific activity `a` as a representative for another agent `ag1` (responsible principal).

5.3.5 Influence — `wasInfluencedBy(id; o2, o1, attrs)`: The capacity of an entity, activity, or agent `o1` (influencer) to have an effect on the character, development, or behavior of another `o2` (influencee) by means of generation, invalidation, communication, derivation, attribution, association, start, end, or delegation.

5.4 Component 4: Bundles, 5.5 Component 5: Alternates & 5.6 Component 6: Collections

Components 4, 5, and 6 provide advanced structural mechanisms for provenance of provenance, cross-context entity linking, and composite data structures:

5.4.1 Bundle Constructor & 5.4.2 Bundle Type (Component 4): A **bundle** is a named set of provenance descriptions, and is itself an entity (with attribute `prov:type='prov:Bundle'`), so allowing provenance of provenance to be expressed. In PROV-N, a bundle is constructed via `bundle id description_1 ... description_n endBundle` inside a `document` container.

5.5.1 Specialization — `specializationOf(infra, supra)` (Component 5): An entity `infra` that is a specialization of another entity `supra` shares all aspects of the latter, and additionally presents more specific aspects of the same thing as the latter. In particular, the lifetime of the entity being specialized contains that of any specialization (e.g., `bbc:news/uk-17595024` is a specialization of the general page `bbc:news/`).

5.5.2 Alternate — `alternateOf(e1, e2)` (Component 5): Two alternate entities `e1` and `e2` present aspects of the same thing. These aspects may be the same or different, and the alternate entities may or may not overlap in time (e.g., a desktop version and a mobile version of the same article).

5.6.1 Collection & 5.6.2 Membership — `hadMember(c, e)` (Component 6): A **collection** is an entity that provides a logical structure to some constituents, which are themselves entities (referred to as its members). PROV-DM defines `prov:Collection` for generic collections, `prov:EmptyCollection` for a collection with no members, and the binary relation `hadMember(c, e)` to assert that entity `e` is a member of collection `c`.

```
      // Component 4 (Bundle), Component 5 (Specialization/Alternate) & Component 6
(Collection) in PROV-N bundle ex:author-view entity(ex:report-v1, [prov:type="report",
ex:version="1"]) specializationOf(ex:report-v1, ex:general-report) alternateOf(ex:report-v1-
pdf, ex:report-v1-html) entity(ex:dataset-c1, [prov:type='prov:Collection'])
hadMember(ex:dataset-c1, ex:sample-e1) endBundle entity(ex:author-view,
[prov:type='prov:Bundle']) wasAttributedTo(ex:author-view, ex:Alice)
```

5.7 Further Elements of PROV-DM (Identifiers, Attributes, Values & Namespaces)

Section 5.7 defines the foundational lexical and syntactic building blocks underlying all PROV-DM concepts: **Identifiers** (5.7.1) are qualified names (`prefix:local`) denoting entities, activities, agents, or relations; **Attributes** (5.7.2) are qualified names paired with values; **Values** (5.7.3) are literals with explicit or implicit XML Schema datatypes (such as strings, integers, `xsd:dateTime`, or `prov:QUALIFIED_NAME`); and **Namespace Declarations** (5.7.4) bind prefixes to IRIs.

Element / Predefined Attribute	Normative Section & Syntax	Normative Definition & Permitted Usage in PROV-DM
`prov:label`	Section 5.7.2.1 `xsd:string`	Provides a human-readable representation of an instance of a PROV-DM type or relation (may be localized via language tags).
`prov:location`	Section 5.7.2.2 Any Value / IRI	An identifiable geographic place (ISO 19112) or non-geographic place (directory, row/column, URI). Applicable to Entity, Activity, Agent, Usage, Generation, Invalidation, Start, and End.
`prov:role`	Section 5.7.2.3 Any Value / QName	The function of an entity or agent with respect to an activity. Applicable to Usage, Generation, Invalidation, Association, Start, and End.
`prov:type`	Section 5.7.2.4 Any Value / QName	Provides further typing information for any construct with an optional set of attribute-value pairs (used for subtyping and extensibility).

6. PROV-DM Extensibility Points & 7. Creating Valid Provenance

6. Extensibility Points: The PROV data model provides extensibility points that allow designers to specialize it for specific applications or domains: (1) Domain-specific *subtypes* of any PROV concept may be declared by asserting `prov:type` with an application namespace qualified name; (2) Domain-specific *attributes* may be added to any extensible PROV-DM expression using custom namespaces; and (3) Domain-specific *roles* (`prov:role`), *locations* (`prov:location`), and *values* may be specified using custom datatypes.

7. Creating Valid Provenance: While PROV-DM defines the conceptual model, expressing provenance about things that change over time requires formal consistency rules. The companion specification **PROV-CONSTRAINTS** defines notions of *event ordering*, *uniqueness*, *impossibility*, and *equivalence/normalization* that provenance descriptions can be checked against in order to validate them and draw sound inferences.

Appendix A. Cross-References to PROV-O and PROV-N (Normative)

Comp	PROV-DM Concept (Sec 5)	PROV-N Notation Expression	PROV-O OWL2 Term
C1	**Entity** (5.1.1)	`entity(id, [attr1=val1, ...])`	`prov:Entity`
C1	**Activity** (5.1.2)	`activity(id, st, et, [attr1=val1, ...])`	`prov:Activity`
C1	**Generation** (5.1.3)	`wasGeneratedBy(id; e, a, t, attrs)`	`prov:wasGeneratedBy / Generation`
C1	**Usage** (5.1.4)	`used(id; a, e, t, attrs)`	`prov:used / prov:Usage`
C1	**Communication** (5.1.5)	`wasInformedBy(id; a2, a1, attrs)`	`prov:wasInformedBy / Communication`
C1	**Start** (5.1.6)	`wasStartedBy(id; a2, e, a1, t, attrs)`	`prov:wasStartedBy / prov:Start`
C1	**End** (5.1.7)	`wasEndedBy(id; a2, e, a1, t, attrs)`	`prov:wasEndedBy / prov:End`
C1	**Invalidation** (5.1.8)	`wasInvalidatedBy(id; e, a, t, attrs)`	`prov:wasInvalidatedBy / Invalidation`
C2	**Derivation** (5.2.1)	`wasDerivedFrom(id; e2, e1, a, g2, u1, attrs)`	`prov:wasDerivedFrom / Derivation`
C2	**Revision / Quotation / PrimarySource** (5.2.2–4)	`wasDerivedFrom(..., [prov:type='prov:Revision' | 'prov:Quotation' | 'prov:PrimarySource'])`	`prov:wasRevisionOf, prov:wasQuotedFrom, prov:hadPrimarySource`
C3	**Agent** & Subtypes (5.3.1)	`agent(id, [prov:type='prov:Person' | ...])`	`prov:Agent, Person, Organization, SoftwareAgent`
C3	**Attribution** (5.3.2)	`wasAttributedTo(id; e, ag, attrs)`	`prov:wasAttributedTo / Attribution`
C3	**Association** & **Plan** (5.3.3)	`wasAssociatedWith(id; a, ag, pl, attrs)`	`prov:wasAssociatedWith / prov:Plan`
C3	**Delegation** (5.3.4)	`actedOnBehalfOf(id; ag2, ag1, a, attrs)`	`prov:actedOnBehalfOf / Delegation`
C3	**Influence** (5.3.5)	`wasInfluencedBy(id; o2, o1, attrs)`	`prov:wasInfluencedBy / Influence`
C4	**Bundle** (5.4.1–5.4.2)	`bundle id ... endBundle / prov:Bundle`	`prov:Bundle`
C5	**Specialization** (5.5.1)	`specializationOf(infra, supra)`	`prov:specializationOf`